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SEMICONDUCTOR DEVICE AND METHOD FOR MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes a first lead, a semiconductor element, a second lead, a conductive member, and a sealing resin. The first lead includes a die pad having a die-pad obverse surface facing a first side in a thickness direction. The semiconductor element has an element obverse surface facing the first side in the thickness direction and includes a first electrode disposed on the element obverse surface. The semiconductor element is mounted on the die-pad obverse surface. The second lead has a second obverse surface facing the first side in the thickness direction and is spaced apart from the first lead in a first direction perpendicular to the thickness direction. The conductive member is electrically bonded to the first electrode and the second obverse surface. The sealing resin covers the semiconductor element. The conductive member is in direct contact with the second lead.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a semiconductor device and a method for manufacturing a semiconductor device.

BACKGROUND ART

[0002] Semiconductor devices incorporating semiconductor elements are provided in various configurations. JP-A-2021-158180 discloses an example of a conventional semiconductor device. The semiconductor device disclosed in JP-A-2021-158180 includes a semiconductor element, a conductive plate, a drive pad, a conductive member, and a sealing resin. The semiconductor element is mounted on the conductive obverse surface of the conductive plate. The conductive member connects an obverse-surface drive electrode, which is disposed on the element obverse surface of the semiconductor element, and the drive pad. The sealing resin seals a portion of the conductive plate, a portion of the drive pad, the semiconductor element, and the conductive member.

[0003] The semiconductor element is bonded to the conductive obverse surface via a conductive bonding material, such as solder. The conductive member is bonded to the obverse-surface drive electrode via a conductive bonding material, such as solder. That is, two layers of a conductive bonding material are present between the conductive obverse surface and the conductive member. The shape of the layers of a conductive bonding material is not consistent, resulting in inconsistencies in the height of the conductive member from the conductive obverse surface (the position of the conductive plate in the thickness direction).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a plan view of a semiconductor device according to a first embodiment of the present disclosure.

[0005] FIG. **2** is a bottom view of the semiconductor device shown in FIG. **1**.

[0006] FIG. **3** is a plan view of the semiconductor device shown in FIG. **1** (with the sealing resin shown as transparent).

[0007] FIG. **4** is a right-side view of the semiconductor device shown in FIG. **1**.
[0008] FIG. **5** is a left-side view of the semiconductor device shown in FIG. **1**.
[0009] FIG. **6** is a sectional view taken along line VI-VI in FIG. **3**.
[0010] FIG. **7** is a sectional view taken along line VII-VII in FIG. **3**.
[0011] FIG. **8** is a sectional view taken along line VIII-VIII in FIG. **3**.
[0012] FIG. **9** is a partially enlarged view of FIG. **6**.
[0013] FIG. **10** is a sectional view for illustrating a step of an example method for manufacturing a semiconductor device shown in FIG. **1**.
[0014] FIG. **11** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **1**.
[0015] FIG. **12** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **1**.
[0016] FIG. **13** is an enlarged sectional view of a semiconductor device according to a first variation of the first embodiment.
[0017] FIG. **14** is an enlarged sectional view of a semiconductor device according to a second embodiment of the present disclosure.
[0018] FIG. **15** is a sectional view of a semiconductor device according to a third embodiment of the present disclosure.
[0019] FIG. **16** is a sectional view of a semiconductor device according to a fourth embodiment of the present disclosure.
[0020] FIG. **17** is a sectional view of a semiconductor device according to a fifth embodiment of the present disclosure.
[0021] FIG. **18** is a sectional view for illustrating a step of an example method for manufacturing a semiconductor device shown in FIG. **17**.
[0022] FIG. **19** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **17**.
[0023] FIG. **20** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **17**.
[0024] FIG. **21** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **17**.
[0025] FIG. **22** is a sectional view of a semiconductor device according to a first variation of the fifth embodiment.
[0026] FIG. **23** is an enlarged sectional view of a semiconductor device according to a second variation of the fifth embodiment.
[0027] FIG. **24** is a sectional view for illustrating a step of an example method for manufacturing a semiconductor device shown in FIG. **23**.
[0028] FIG. **25** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **23**.
[0029] FIG. **26** is a sectional view for illustrating a step of the example method for manufacturing a semiconductor device shown in FIG. **23**.

DETAILED DESCRIPTION OF EMBODIMENTS

[0030] The following specifically describes preferred embodiments of the present disclosure with reference to the drawings.
[0031] In the present disclosure, the terms such as “first”, “second”, “third”, and so on are used only as labels and not to imply an order of the items referred to by the terms.
[0032] In the description of the present disclosure, the expression “An object A is formed in an object B”, and “An object A is formed on an object B” imply the situation where, unless otherwise specifically noted, “the object A is formed directly in or on the object B”, and “the object A is formed in or on the object B, with something else interposed between the object A and the object B”. Likewise, the expressions “An object A is arranged in an object B”, and “An object A is

arranged on an object B” imply the situation where, unless otherwise specifically noted, “the object A is arranged directly in or on the object B”, and “the object A is arranged in or on the object B, with something else interposed between the object A and the object B”. Further, the expression “An object A is located on an object B” implies the situation where, unless otherwise specifically noted, “the object A is located on the object B, in contact with the object B”, and “the object A is located on the object B, with something else interposed between the object A and the object B”. Still further, the expression “An object A overlaps with an object B as viewed in a certain direction” implies the situation where, unless otherwise specifically noted, “the object A overlaps with the entirety of the object B”, and “the object A overlaps with a part of the object B”.

First Embodiment

[0033] With reference to FIGS. **1** to **9**, the following describes a semiconductor device A**10** according to a first embodiment of the present disclosure. The semiconductor device A**10** includes a plurality of leads **1**A, **1**B, and **1**C, a semiconductor element **2**, an insulating part **3**, a metal laminate **4**, a conductive member **5**, conductive bonding materials **61** and **62**, and a sealing resin **7**.

[0034] FIG. **1** is a plan view of the semiconductor device A**10**. FIG. **2** is a bottom view of the semiconductor device A**10**. FIG. **3** is a plan view of the semiconductor device A**10**. FIG. **4** is a right-side view of the semiconductor device A**10**. FIG. **5** is a left-side view of the semiconductor device A**10**. FIG. **6** is a sectional view taken along line VI-VI in FIG. **3**. FIG. **7** is a sectional view taken along line VII-VII in FIG. **3**. FIG. **8** is a sectional view taken along line VIII-VIII in FIG. **3**. For convenience of description, FIG. **3** shows the sealing resin **7** as transparent.

[0035] In the description of the semiconductor device A**10**, the thickness direction of the semiconductor element **2** is referred to as “thickness direction z”. A direction perpendicular to the thickness direction z is referred to as “first direction x”. The direction perpendicular to both the thickness direction z and the first direction x is referred to as “second direction y”. As shown in FIGS. **1** and **2**, the semiconductor device A**10** is substantially rectangular as viewed in the thickness direction z. The size of the semiconductor device A**10** is not specifically limited.

[0036] The leads **1**A, **1**B, and **1**C are formed from a metal plate (a lead frame) through punching, bending, and other processes. The constituent material of the leads **1**A, **1**B, and **1**C is not specifically limited, and examples of the constituent material include copper (Cu), nickel (Ni), and alloys of these metals. In the present embodiment, the constituent material of the leads **1**A, **1**B, and **1**C is Cu. The thickness of the leads **1**A, **1**B, and **1**C may be, but not limited to, 0.1 to 0.3 mm, for example.

[0037] As shown in FIG. **3**, the lead **1**A is located on a first side in the first direction x from the leads **1**B and **1**C. The leads **1**B and **1**C are arranged next to each other in the second direction y. As viewed in the thickness direction z, the leads **1**A to **1**C are spaced apart from each other. In terms of the size as viewed in the thickness direction z, the lead **1**A is the largest, and the lead **1**C is the smallest.

[0038] As shown in FIG. **3** and FIGS. **6** to **8**, the lead **1**A includes a die pad **12** and a plurality of (four in the present embodiment) first terminal portions **13**. For example, the die pad **12** is rectangular as viewed in the thickness direction z. The die pad **12** has an obverse surface **121** and a reverse surface **122**. The obverse surface **121** faces a first side in the thickness direction z, and the reverse surface **122** faces away from the obverse surface **121** (faces a second side in the thickness direction z). The obverse surface **121** is where the semiconductor element **2** is mounted. As shown in, for example, FIGS. **2** and **6**, the reverse surface **122** is exposed from the sealing resin **7**. When the semiconductor device A**10** is mounted on a circuit board that is not shown in the figures, the reverse surface **122** is joined to the circuit board using a bonding material, such as solder.

[0039] The first terminal portions **13** are located on the first side in the first direction x from the die pad **12** (the right side in FIG. **6**). Each first terminal portion **13** is connected to the end of the die pad **12** on the first side in the first direction x and extends toward the first side in the first direction x. The first terminal portions **13** are arranged at intervals in the second direction y. Each first terminal portion **13** includes a reverse-surface mounting portion **131**. The reverse-surface mounting portion **131** faces the second side in the thickness direction z (the bottom side in FIG. **6**). The reverse-surface mounting portion **131** is exposed from the sealing resin **7**. When the semiconductor device A**10** is mounted on a circuit board that is not shown in the figures, the reverse-surface mounting portion **131** is joined to the circuit board using a bonding material, such as solder.

[0040] As shown in FIGS. **3** and **6**, the lead **1**B includes a pad portion **14**, a plurality of (three in the present embodiment) second terminal portions **15**, and a plurality of (three in the present embodiment) bent portions **16**. The pad portion **14** is located on the first side in the thickness direction z (the upper side in FIG. **6**) from the second terminal portions **15**. In the first direction x, the pad portion **14** is located inward from the second terminal portions **15** and is covered with the sealing resin **7**. The pad portion **14** has an obverse surface **141** facing the first side in the thickness direction z.

[0041] The second terminal portions **15** are located on the second side in the first direction x from the die pad **12** of the lead **1**A (the left side in FIG. **6**). Each second terminal portion **15** extends toward the second side in the first direction x. The second terminal portions **15** are arranged at intervals in the second direction y. Each second terminal portion **15** includes a reverse-surface mounting portion **151**. The reverse-surface mounting portion **151** faces the second side in the thickness direction z (the bottom side in FIG. **6**). The reverse-surface mounting portion **151** is exposed from the sealing resin **7**. When the semiconductor device A**10** is mounted on a circuit board that is not shown in the figures, the reverse-surface mounting portion **151** is joined to the circuit board using a bonding material, such as solder. Each bent portion **16** connects the pad portion **14** and a corresponding second terminal portion **15** and is bent as viewed in the second direction y.

[0042] As shown in FIGS. **3** and **7**, the lead **1**C includes a pad portion **17**, a second terminal portion **18**, and a bent portion **19**. The pad portion **17** is located on the first side in the thickness direction z (the upper side in FIG. **7**) from the second terminal portion **18**. In the first direction x, the pad portion **17** is located inward from the second terminal portion **18** and is covered with the sealing resin **7**.

[0043] The second terminal portion **18** is located on the second side in the first direction x (the left side in FIG. **7**) from the die pad **12** of the lead **1**A. The second terminal portion **18** extends toward the second side in the first direction x. The second terminal portions **15** of the lead **1**B and the second terminal portion **18** of the lead **1**C are arranged at intervals in the second direction y. The second terminal portion **18** has a reverse-surface mounting portion **181**. The reverse-surface mounting portion **181** faces the second side in the thickness direction z (the bottom side in FIG. **7**). The reverse-surface mounting portion **181** is exposed from the sealing resin **7**. When the semiconductor device A**10** is mounted on a circuit board that is not shown in the figures, the reverse-surface mounting portion **181** is joined to the circuit board using a bonding material, such as solder. The bent portion **19** connects the pad portion **17** and the second terminal portion **18** and is bent as viewed in the second direction y.

[0044] The semiconductor element **2** implements the electrical function of the semiconductor device A**10**. The type of the semiconductor element **2** is not specifically limited. In the present embodiment, the semiconductor element **2** is configured as a transistor. The semiconductor element **2** is mounted on the obverse surface **121** of the die pad **12**. As shown in FIGS. **3** and **6** to **8**, the semiconductor element **2** includes an element body **20**, a first electrode **21**, a second electrode **22**, and a third electrode **23**.

[0045] The element body **20** is rectangular as viewed in the thickness direction z. The element body **20** has an element obverse surface **201** and an element reverse surface **202**. The element obverse surface **201** and the element reverse surface **202** face away from each other in the thickness direction z. The element obverse surface **201** faces the same side as the obverse surface **121** of the die pad **12** in the thickness direction z (the first side in the thickness direction z). Thus, the element reverse surface **202** faces the obverse surface **121**.

[0046] The first electrode **21** and the third electrode **23** are disposed on the element obverse surface **201**. The second electrode **22** is disposed on the element reverse surface **202**. The constituent materials of first electrode **21**, the second electrode **22**, and the third electrode **23** include copper, aluminum (Al), and alloys of these metals. In the present embodiment, the first electrode **21** is the source electrode, the second electrode **22** is the drain electrode, and the third electrode **23** is the gate electrode.

[0047] In the present embodiment, the first electrode **21** covers most of the element obverse surface **201**. Specifically, the first electrode **21** covers the element obverse surface **201**, which is rectangular, except for its periphery and one corner (the lower right corner in FIG. **3**). The first electrode **21** includes a first-electrode pad portion **212**. As viewed in the thickness direction z, the first-electrode pad portion **212** is located inside the insulating part **3**. The third electrode **23** is located at one corner of the element obverse surface **201** (the lower right corner in FIG. **3**). The second electrode **22** covers the entire (or substantially entire) element reverse surface **202**.

[0048] The second electrode **22** is bonded to the obverse surface **121** of the die pad **12** via a conductive bonding material **62**. The conductive bonding material **62** electrically connects the die pad **12** and the second electrode **22**. The conductive bonding material **62** is solder, for example.

[0049] The semiconductor device A**10** includes a wire **65**. The wire **65** is electrically bonded to the third electrode **23** and the pad portion **17** of the lead **1**C. The wire **65** electrically connects the third electrode **23** and the lead **1**C.

[0050] As shown in FIGS. **3** and **6** to **8**, the insulating part **3** is arranged to extend across the first electrode **21** and the element obverse surface **201**. As viewed in the thickness direction z, the insulating part **3** has an annular shape that overlaps with the outer edge of the first electrode **21**. The outer edge of the insulating part **3** is located closely along to the outer edge of the element obverse surface **201** as viewed in the thickness direction z. As viewed in the thickness direction z, the region of the first electrode **21** located inside the inner edge of the insulating part **3** is the first-electrode pad portion **212**. In one example, the insulating part **3** is composed of a plurality of insulating layers stacked on top of each other. The insulating part **3** may include a lower insulating layer made of nitride, and an upper insulating layer made of a resin material, for example. Examples of nitride forming the lower insulating layer include SiN, SiON, and SiO.sub.2. Examples of the resin material forming the upper insulating layer include a polyimide resin.

[0051] As shown in FIGS. **3** and **6** to **8**, the metal laminate **4** is arranged to extend across the first electrode **21** and the insulating part **3** and is composed of a plurality of metal layers that are stacked on top of each other, for example. The metal laminate **4** may include a metal layer containing titanium (Ti), a metal layer containing nickel, and a metal layer containing silver (Ag), and they are stacked in the stated order. Unlike the present embodiment, the semiconductor device of the present disclosure may not include the insulating part **3** and the metal laminate **4**.

[0052] As shown in FIGS. **3** and **6**, the conductive member **5** is electrically bonded to the first electrode **21** of the semiconductor element **2** and the lead **1**B. The conductive member **5** is made of a metal plate having varying thicknesses (having multi-gauge strips). The metal may be copper (Cu) or a copper alloy. In the present embodiment, the constituent material of the conductive member **5** is the same as that of the lead **1**B, namely Cu. The conductive member **5** is a metal plate having been processed by bending and punching. In the present embodiment, the conductive member **5** includes an element bonding portion **51**, a lead bonding portion **52**, and an intermediate portion **53**. The element bonding portion **51** is a portion with a greater thickness (the dimension in the thickness direction z) among the multi-gauge strips and has a rectangular shape that is elongated in the second direction y as viewed in the thickness direction z. The element bonding portion **51** is electrically bonded to the first-electrode pad portion **212** of the first electrode **21** via the conductive bonding material **61**. The conductive bonding material **61** electrically connects the element bonding portion **51** (the conductive member **5**) and the first-electrode pad portion **212**. The conductive bonding material **61** is solder, for example.

[0053] The element bonding portion **51** has an obverse surface **511**, a reverse surface **512**, and an end surface **513**. The obverse surface **511** and the reverse surface **512** face away from each other in the thickness direction z. The obverse surface **511** faces the same side as the obverse surface **121** of the die pad **12** in the thickness direction z (the first side in the thickness direction z). As shown in FIGS. **1** and **6** to **8**, the obverse surface **511** is exposed from the sealing resin **7**. The reverse surface **512** faces the same side as the reverse surface **122** of the die pad **12** in the thickness direction z (the second side in the thickness direction z). As shown in FIGS. **6** to **8**, the reverse surface **512** is bonded to the first-electrode pad portion **212** of the semiconductor element **2**. This allows the conductive member **5** to dissipate heat generated by the semiconductor element **2** through its obverse surface **511**. The end surface **513** is connected to the obverse surface **511** and the reverse surface **512** and is located between the obverse surface **511** and the reverse surface **512** in the thickness direction z. The end surface **513** faces the first side in the first direction x. The shape of the element bonding portion **51** is not specifically limited.

[0054] The lead bonding portion **52** is electrically bonded to the pad portion **14** of the lead **1**B. The lead bonding portion **52** is bonded in direct contact with the obverse surface **141** of the pad portion **14**. In the present embodiment, the lead bonding portion **52** is joined to the obverse surface **141** of the pad portion **14** by ultrasonic bonding. As shown in FIG. **6**, the lead bonding portion **52** is appropriately bent as viewed in the second direction y, forming a protruding portion **521** that is closer to the second side in the thickness direction z (lower side in the figure) than the rest. A solid-state bonding interface **59** is present between the protruding portion **521** and the pad portion **14**. The solid-state bonding interface **59** forms when the protruding portion **521** and the pad portion **14** are joined together in the solid state using ultrasonic vibrations and pressure applied during ultrasonic bonding process. Alternatively, other solid-state bonding methods, including diffusion bonding and thermal compression bonding, may be used to join the lead bonding portion **52** to the obverse surface **141** of the pad portion **14**.

[0055] The intermediate portion **53** is present between the element bonding portion **51** and the lead bonding portion **52** in the first direction x. The intermediate portion **53** is connected to both the element bonding portion **51** and the lead bonding portion **52**.

[0056] The sealing resin **7** covers the semiconductor element **2**, the insulating part **3**, and the metal laminate **4**, as well as a portion of each of the conductive member **5** and the leads **1**A, **1**B, and **1**C. The sealing resin **7** is made of a black epoxy resin, for example.

[0057] As shown in FIGS. **1**, **2**, and **4** to **8**, the sealing resin **7** has a resin obverse surface **71**, a resin reverse surface **72**, and resin side surfaces **73** to **76**. The resin obverse surface **71** and the resin reverse surface **72** face away from each other in the thickness direction z. The resin obverse surface **71** faces the first side in the thickness direction z, similarly to the obverse surface **121** of the element obverse surface **201**. As shown in FIG. **1**, the resin obverse surface **71** exposes the obverse surface **511** of the element bonding portion **51** of the conductive member **5**. The resin reverse surface **72** faces the second side in the thickness direction z, similarly to the element reverse surface **202** and the reverse surface **122**. As shown in FIG. **2**, the resin reverse surface **72** exposes the reverse surface **122** of the die pad **12**, the reverse-surface mounting portions **131** of the first terminal portions **13**, the reverse-surface mounting portions **151** of the second terminal portions **15**, and the reverse-surface mounting portion **181** of the second terminal portion **18**.

[0058] The resin side surfaces **73** to **76** are each connected to the resin obverse surface **71** and the resin reverse surface **72**, and each located between the resin obverse surface **71** and the resin reverse surface **72** in the thickness direction z. The resin side surfaces **73** and **74** face away from each other in the first direction x. The resin side surface **73** faces the first side in the first direction x, and the resin side surface **74** faces the second side in the first direction x. The resin side surfaces **75** and **76** face away from each other in the second direction y. The resin side surface **75** faces a first side in the second direction y, and the resin side surface **76** faces a second side in the second direction y. As shown in FIG. **1**, a portion of each first terminal portion **13** protrudes from the resin side surface **73**. Also, a portion of each second terminal portion **15** and a portion of the second terminal portion **18** protrude from the resin side surface **74**. In the illustrated example, the resin side surfaces **73** to **76** are each slightly inclined relative to the thickness direction z. Note that the shape of the sealing resin **7** shown in FIGS. **1**, **2**, and **4** to **8** is only one example. The shape of the sealing resin **7** is not limited to this example.

[0059] The following describes a method for manufacturing a semiconductor device A**10**, with reference to FIGS. **10** to **12**. FIGS. **10** to **12** are sectional views corresponding to FIG. **6**, each illustrating a step of the method for manufacturing a semiconductor device A**10**.

[0060] First, a lead frame **100** and a semiconductor element **2** are prepared as shown in FIG. **10**. The lead frame **100** is a plate from which the leads **1**A, **1**B, and **1**C will be formed. The lead frame **100** is formed from a metal plate through punching, bending, and other processes. The process for forming the lead frame **100** is not specifically limited. The description of the process for manufacturing the semiconductor element **2** is omitted. In a separate step, a conductive member **5** is prepared. The conductive member **5** is formed from a metal plate having multi-gauge strips through punching, bending, and other processes. The process for forming the conductive member **5** is not specifically limited.

[0061] Subsequently, as shown in FIG. **10**, a solder paste **60** is applied to the obverse surface **101** of the lead frame **100** to cover the portion for forming the obverse surface **121** of the die pad **12**. Then, the semiconductor element **2** is placed on the applied solder paste **60**.

[0062] Subsequently, as shown in FIG. **11**, the solder paste **60** is applied to the first electrode **21** of the semiconductor element **2**. Then, as shown in FIG. **11**, the conductive member **5** is placed to span across the semiconductor element **2** and the portion of the lead frame **100** that is for forming the pad portion **14**. In this state, the element bonding portion **51** is placed on the solder paste **60**, whereas the protruding portion **521** of the lead bonding portion **52** is placed directly on the portion of the lead frame **100** that is for forming the pad portion **14**.

[0063] Subsequently, the protruding portion **521** of the lead bonding portion **52** is joined by ultrasonic bonding to the portion of the lead frame **100** that is for forming the pad portion **14**. Specifically, the protruding portion **521** is placed into direct contact with the portion for forming the pad portion **14** and pressed against it. In this state, ultrasonic vibrations are applied to create a solid-state bond. As a result, a solid-state bonding interface **59** forms between the protruding portion **521** and the portion for forming the pad portion **14** as shown in FIG. **12**.

[0064] Subsequently, a reflow process is performed. The reflow process involves melting the solder paste **60** and subsequent cooling, allowing the molten solder to solidify. This forms a conductive bonding material **62**, bonding the semiconductor element **2** to the portion of the lead frame **100** that is for forming the die pad **12**. This also forms a conductive bonding material **61**, bonding the element bonding portion **51** of the conductive member **5** to the first electrode **21**.

[0065] Subsequently, a wire **65** is bonded using wire bonding. Then, a sealing resin **7** is formed by molding so as to cover the semiconductor element **2**, the insulating part **3**, and the metal laminate **4**, as well as a portion of each of the conductive member **5** and the lead frame **100**. Subsequently, the lead frame **100** is appropriately cut, separating the leads **1**A, **1**B and **1**C from each other. Through the steps described above, the semiconductor device A**10** as shown in FIGS. **1** to **9** is manufactured.

[0066] The following describes operation of the present embodiment.

[0067] According to the present embodiment, the lead bonding portion **52** of the conductive member **5** is bonded in direct contact with the obverse surface **141** of the pad portion **14**. That is, no bonding material or the like is interposed between the lead bonding portion **52** and the obverse surface **141**, and the height (the position in the thickness direction z) of the conductive member **5** from the obverse surface **121** of the die pad **12** is determined by the height of the obverse surface **141** of the pad portion **14**. The semiconductor device A**10** thus ensures that the height of the conductive member **5** from the obverse surface **121** is controlled to be consistent, irrespective of the thicknesses (the dimensions in the thickness direction z) of the conductive bonding materials **61** and **62**. By controlling the position of the conductive member **5** to an appropriate height, the sealing resin **7** is prevented from forming on the obverse surface **511** of the element bonding portion **51**. In addition, the lead bonding portion **52** is joined to the pad portion **14** before the reflow process. This helps prevent the conductive member **5** from rotating around an axis extending in the thickness direction z when the solder paste **60** melts during the reflow process.

[0068] According to the present embodiment, in addition, the lead bonding portion **52** of the conductive member **5** is joined to the obverse surface **141** of the pad portion **14** by ultrasonic bonding. Thus, the lead bonding portion **52** and the pad portion **14** are bonded in direct contact, without any bonding material interposed between them.

[0069] According to the present embodiment, in addition, the conductive member **5** is made of the same constituent material as that of the lead **1**B, namely Cu. This allows the lead bonding portion **52** and the pad portion **14** to be firmly bonded by ultrasonic bonding.

[0070] According to the present embodiment, in addition, the obverse surface **511** of the element bonding portion **51** of the conductive member **5** is exposed from the resin obverse surface **71**. This allows the semiconductor device A**10** to dissipate heat generated by the semiconductor element **2** through the obverse surface **511** of the conductive member **5**. The reverse surface **122** of the die pad **12** is exposed from the resin reverse surface **72**. This allows the semiconductor device A**10** to dissipate heat generated by the semiconductor element **2** through the reverse surface **122** of the die pad **12**. That is, the semiconductor device A**10** is configured to dissipate heat from both sides in the thickness direction z, which is more efficient than dissipating heat from only one side in the thickness direction z.

[0071] FIG. **13** shows a variation of the semiconductor device A**10** according to the first embodiment. In the figure, elements that are identical or similar to those of the embodiment described above are indicated by the same reference numerals, and redundant descriptions are omitted.

First Variation of First Embodiment:

[0072] FIG. **13** is a semiconductor device A**11** according to a first variation of the first embodiment. FIG. **13** is a partially enlarged sectional view of the semiconductor device A**11** and corresponds to FIG. **9**. The pad portion **14** of the lead **1**B according to this variation includes a plating layer **142** on the obverse surface **141**. The constituent material of the plating layer **142** is silver (Ag), for example, but not limited to this. In addition, the lead bonding portion **52** of the conductive member **5** according to this variation includes a plating layer **522** on a contact surface **521***a* of the protruding portion **521**, facing the second side in the thickness direction z. The constituent material of the plating layer **522** is the same as that of the plating layer **142**, which is silver (Ag) in this variation. When created by ultrasonic bonding, silver-to-silver bonds are stronger than silver-to-copper bonds. Hence, the semiconductor device A**11** forms a stronger bond between the lead bonding portion **52** and the pad portion **14** than the semiconductor device A**10**.

[0073] FIGS. **14** to **26** show other embodiments of the present disclosure. In these figures, elements that are identical or similar to those of the embodiment described above are indicated by the same reference numerals.

Second Embodiment

[0074] FIG. **14** is a view for illustrating a semiconductor device A**20** according to a second embodiment of the present disclosure. FIG. **14** is an enlarged sectional view of the semiconductor device A**20** and corresponds to FIG. **9**. The semiconductor device A**20** of the present embodiment differs from the first embodiment in the process used for bonding the lead bonding portion **52** of the conductive member **5** and the pad portion **14** of the lead **1**B. Other than that, the configurations and operation of the present embodiment are similar to those of the first embodiment. Note that the present embodiment may be combined with any part of the first embodiment and the variations described above.

[0075] For the semiconductor device A**20** of the present embodiment, the lead bonding portion **52** of the conductive member **5** and the pad portion **14** of the lead **1**B are joined by laser welding using a laser beam. The lead bonding portion **52** has a welding mark **523** that extends to the interior of the pad portion **14**. The welding mark **523** forms when a portion of the lead bonding portion **52** and a portion of the pad portion **14** are fused and joined together during laser welding. Note that the lead bonding portion **52** may have a plurality of welding marks **523**.
[0076] According to the present embodiment, the lead bonding portion **52** of the conductive member **5** is bonded in direct contact with the obverse surface **141** of the pad portion **14**. Thus, similarly to the semiconductor device A**10**, the semiconductor device A**20** ensures that the height of the conductive member **5** from the obverse surface **121** of the die pad **12** is controlled to be consistent. According to the present embodiment, in addition, the lead bonding portion **52** of the conductive member **5** is joined to the obverse surface **141** of the pad portion **14** by laser welding. Thus, the lead bonding portion **52** and the pad portion **14** are bonded in direct contact, without any bonding material interposed between them. In addition, the semiconductor device A**20** has a configuration in common with the semiconductor device A**10**, thereby achieving the same effect as the semiconductor device A**10**.
[0077] As can be understood from the first and second embodiments, the process for joining the lead bonding portion **52** of the conductive member **5** and the pad portion **14** of the lead **1**B is not specifically limited. It is sufficient that the lead bonding portion **52** and the pad portion **14** are bonded in direct contact with each other.
Third Embodiment
[0078] FIG. **15** is a view for illustrating a semiconductor device A**30** according to a third embodiment of the present disclosure. FIG. **15** is a sectional view of the semiconductor device A**30** and corresponds to FIG. **6**. The semiconductor device A**30** of the present embodiment differs from the first embodiment in the addition of a heat-conducting member **9** that is exposed from the resin obverse surface **71** of the sealing resin **7**. Other than that, the configurations and operation of the present embodiment are similar to those of the first embodiment. Note that the present embodiment may be combined with any part of the first and second embodiments and the variations described above.
[0079] The semiconductor device A**30** of the present embodiment additionally includes the heat-conducting member **9**. The heat-conducting member **9** includes an insulating plate **9***a* and two metal layers **9***b*. The insulating plate **9***a* is a plate having, for example, a rectangular shape as viewed in the thickness direction z. The insulating plate **9***a* is made of a ceramic material with excellent thermal conductivity, which is aluminum nitride (AlN) in the present embodiment. The shape and the constituent material of the insulating plate **9***a* are not specifically limited. The two metal layers **9***b* are formed on the opposite surfaces of the heat-conducting member **9** in the thickness direction z. Each metal layer **9***b* is identical in shape and size to the insulating plate **9***a* as viewed in the thickness direction z. The constituent material of each metal layer **9***b* is not specifically limited, and examples of the constituent material include copper (Cu), silver (Ag), gold (Au), and alloys of these metals. In the present embodiment, the metal layers **9***b* are made of copper (Cu). In the present embodiment, the heat-conducting member **9** is a direct bonded copper (DBC) substrate. A DBC substrate is a ceramic substrate with silver foil bonded to both sides.
[0080] The heat-conducting member **9** has an obverse surface **91** and a reverse surface **92**. The obverse surface **91** and the reverse surface **92** face away from each other in the thickness direction z. The obverse surface **91** faces the first side in the thickness direction z, and the reverse surface **92** faces away from the obverse surface **91** (faces the second side in the thickness direction z). The reverse surface **92** of the heat-conducting member **9** is bonded to the obverse surface **511** of the element bonding portion **51** of the conductive member **5**. The obverse surface **91** of the heat-conducting member **9** is exposed from the sealing resin **7**.
[0081] Note that the heat-conducting member **9** is not limited to a DBC substrate. For example, the

heat-conducting member **9** may be a direct plated copper (DPC) substrate, which consists of a ceramic plate with copper plating applied to both sides. The heat-conducting member **9** may be a plating layer made of copper, for example, or a thermal conductive material, such as a thermal interface material (TIM).

[0082] According to the present embodiment, the lead bonding portion **52** of the conductive member **5** is bonded in direct contact with the obverse surface **141** of the pad portion **14**. Thus, similarly to the semiconductor device A**10**, the semiconductor device A**30** ensures that the height of the conductive member **5** from the obverse surface **121** of the die pad **12** is controlled to be consistent. In addition, according to the present embodiment, the obverse surface **91** of the heat-conducting member **9** is exposed from the sealing resin **7**. This allows the semiconductor device A**30** to dissipate heat generated by the semiconductor element **2** through the obverse surface **91** of the heat-conducting member **9** via the conductive member **5**. In addition, the semiconductor device A**30** has a configuration in common with the semiconductor device A**10**, thereby achieving the same effect as the semiconductor device A**10**.

Fourth Embodiment

[0083] FIG. **16** is a view for illustrating a semiconductor device A**40** according to a fourth embodiment of the present disclosure. FIG. **16** is a sectional view of the semiconductor device A**40** and corresponds to FIG. **6**. The semiconductor device A**40** of the present embodiment differs from the first embodiment in that the conductive member **5** is covered with the sealing resin **7** and is not exposed from the resin obverse surface **71**. Other than that, the configurations and operation of the present embodiment are similar to those of the first embodiment. Note that the present embodiment may be combined with any part of the first to third embodiments and the variations described above.

[0084] The semiconductor device A**40** of the present embodiment includes the conductive member **5** that is entirely covered with the sealing resin **7**. Hence, the obverse surface **511** of the element bonding portion **51** is not exposed from the resin obverse surface **71** of the sealing resin **7**. In addition, the semiconductor device A**40**, unlike the semiconductor device A**30**, does not include the heat-conducting member **9**.

[0085] According to the present embodiment, the lead bonding portion **52** of the conductive member **5** is bonded in direct contact with the obverse surface **141** of the pad portion **14**. Thus, similarly to the semiconductor device A**10**, the semiconductor device A**40** ensures that the height of the conductive member **5** from the obverse surface **121** of the die pad **12** is controlled to be consistent. In addition, the semiconductor device A**40** has a configuration in common with the semiconductor device A**10**, thereby achieving the same effect as the semiconductor device A**10**.

Fifth Embodiment

[0086] FIGS. **17** to **21** are views for illustrating a semiconductor device A**50** according to a fifth embodiment of the present disclosure. FIG. **17** is a sectional view of the semiconductor device A**50** and corresponds to FIG. **6**. FIGS. **18** to **21** are sectional views each for illustrating a step of an example method for manufacturing the semiconductor device A**50**. The semiconductor device A**50** of the present embodiment differs from the first embodiment in the addition of a positioning member **8** that determines the height of the conductive member **5** from the obverse surface **121** of the die pad **12**. Other than that, the configurations and operation of the present embodiment are similar to those of the first embodiment. Note that the present embodiment may be combined with any part of the first to fourth embodiments and the variations described above.

[0087] The semiconductor device A**50** of the present embodiment includes the positioning member **8**. The positioning member **8** is made of an insulating material and is placed in contact with the conductive member **5** and the obverse surface **121** of the die pad **12**. The positioning member **8** is made of a synthetic resin, for example. The type of synthetic resin is not specifically limited. The positioning member **8** is located on the opposite side of the semiconductor element **2** from the lead **1**B in the first direction x.

[0088] As viewed in the second direction y, the positioning member **8** has an L shape and includes a first portion **81** and a second portion **82**. The second portion **82** is formed with a plate extending in the first direction x, and its end surface **82***a* facing the second side in the first direction x is bonded in contact with the end surface **513** of the element bonding portion **51**. The process for the bonding is not limited, and examples of the process include thermal compression bonding, which involves heating the positioning member **8** and the conductive member **5** and pressing them together to form the bond. Alternatively, the positioning member **8** may be formed in contact with the end surface **513** of the element bonding portion **51** by introducing molten resin material into a mold and then allowing it to harden.
[0089] The first portion **81** is a plate extending in the thickness direction z, and its end on the first side in the thickness direction z is connected to the end of the second portion **82** on the first side in the first direction x. The end surface **81***a* of the first portion **81** facing the second side in the thickness direction z is in contact with the obverse surface **121** of the die pad **12**. As will be described later regarding the manufacturing method, the positioning member **8** and the conductive member **5** are first joined into a single unit. After that, the conductive member **5** is placed so that the end surface **81***a* of the first portion **81** is in contact with the obverse surface **121** of the die pad **12**. Then, the lead bonding portion **52** is bonded to the pad portion **14**, and the conductive bonding materials **61** and **62** are hardened. As a result, the end surface **81***a* of the first portion **81** is fixed in contact with the obverse surface **121** of the die pad **12**.
[0090] The material of the positioning member **8** is not limited to synthetic resin, and other insulating materials are also usable. For example, the positioning member **8** may be made of a ceramic material. Additionally, the shape and position of the positioning member **8** are not specifically limited either. The semiconductor device A**50** may include a plurality of positioning members **8**. In such a case, the arrangement of the positioning members **8** is not specifically limited.
[0091] The following describes a method for manufacturing the semiconductor device A**50**, with reference to FIGS. **18** to **21**. FIGS. **18** to **21** are sectional views corresponding to FIG. **6**, each illustrating a step of a method for manufacturing the semiconductor device A**50**.
[0092] First, a conductive member **5** and a positioning member **8** are prepared as shown in FIG. **18**. For example, the positioning member **8** may be formed by injection molding using a mold. Note, however, that the process for forming the positioning member **8** is not specifically limited.
[0093] Subsequently, as shown in FIG. **18**, the positioning member **8** is joined to conductive member **5** by thermal compression bonding. Specifically, the positioning member **8** and the conductive member **5** are heated to an appropriate temperature. Then, a pressure is applied to bring the end surface **82***a* into intimate contact with the end surface **513**. This causes the positioning member **8** to undergo plastic deformation to form the bond. Through this process, the positioning member **8** and the conductive member **5** are joined into a single unit. Alternatively, the positioning member **8** may be formed in contact with the end surface **513** of the conductive member **5** by placing the conductive member **5** in a mold, introducing molten resin material into the mold, and then allowing it to harden.
[0094] Subsequently, as in the first embodiment, the lead frame **100** and the semiconductor element **2** are prepared, solder paste **60** is applied to the obverse surface **101** of the lead frame **100** to cover the portion for forming the obverse surface **121** of the die pad **12**, and the semiconductor element **2** is placed on the applied solder paste **60** (see FIG. **10**). Subsequently, as shown in FIG. **19**, the solder paste **60** is applied to the first electrode **21** of the semiconductor element **2**.
[0095] Subsequently, as shown in FIG. **20**, the conductive member **5** is placed to span across the semiconductor element **2** and the portion of the lead frame **100** that is for forming the pad portion **14**. In this state, the element bonding portion **51** is placed on the solder paste **60**, whereas the protruding portion **521** of the lead bonding portion **52** is placed directly on the portion of the lead frame **100** that is for forming the pad portion **14**. In addition, the conductive member **5** is placed

such that the end surface **81***a* of the positioning member **8**, which has been integrated with the conductive member **5**, is in contact with the portion of the obverse surface **101** of the lead frame **100** that is for forming the obverse surface **121** of the die pad **12**.

[0096] Subsequently, the protruding portion **521** of the lead bonding portion **52** is joined to the portion of the lead frame **100** that is for forming the pad portion **14** by ultrasonic bonding. As a result, a solid-state bonding interface **59** forms between the protruding portion **521** and the portion for forming the pad portion **14** as shown in FIG. **21**. The subsequent steps are the same as those of the first embodiment.

[0097] According to the present embodiment, the lead bonding portion **52** of the conductive member **5** is bonded in direct contact with the obverse surface **141** of the pad portion **14**. Thus, similarly to the semiconductor device A**10**, the semiconductor device A**50** ensures that the height of the conductive member **5** from the obverse surface **121** of the die pad **12** is controlled to be consistent. In addition, the semiconductor device A**50** has a configuration in common with the semiconductor device A**10**, thereby achieving the same effect as the semiconductor device A**10**.

[0098] Additionally, the semiconductor device A**50** of the present embodiment includes the positioning member **8**. The positioning member **8** is integrated with the conductive member **5** by bonding the end surface **82***a* of the second portion **82** directly to the end surface **513** of the element bonding portion **51**. The conductive member **5** is secured in the state where the end surface **81***a* of the positioning member **8**, which has been integrated with the conductive member **5**, is in contact with the obverse surface **121** of the die pad **12**. Thus, the height (the position in the thickness direction z) of the conductive member **5** from the obverse surface **121** is determined also by the dimension of the positioning member **8** in the thickness direction z. This configuration ensures that the height of the element bonding portion **51** is more accurately determined than a configuration without the positioning member **8**. The semiconductor device A**50** thus provides more precise control over the height of the conductive member **5** from the obverse surface **121**.

[0099] According to the present embodiment, in addition, the positioning member **8** includes the first portion **81** extending in the thickness direction z, and the second portion **82** extending in the first direction x. This allows the positioning member **8** to be in contact with the end surface **513** of the element bonding portion **51** at the end surface **82***a* of the second portion **82** and with the obverse surface **121** of the die pad **12** at the end surface **81***a* of the first portion **81**.

[0100] According to the present embodiment, in addition, the positioning member **8** is located on the opposite side of the semiconductor element **2** from the lead **1**B in the first direction x. The lead bonding portion **52** of the conductive member **5** is bonded to the pad portion **14** of the lead **1**B. That is, the height of the conductive member **5** is determined on the first side of the semiconductor element **2** in the first direction x by the positioning member **8**, and on the second side of the semiconductor element **2** in the first direction x by the height of the obverse surface **141** of the pad portion **14**. In other words, the height of the semiconductor device A**50** is determined on both sides of the semiconductor element **2** in the first direction x, ensuring more accurate control of the height of the conductive member **5** compared to configurations with the positioning member **8** placed differently.

[0101] FIGS. **22** to **26** show variations of the semiconductor device A**50** according to the fifth embodiment. In these figures, elements that are identical or similar to those of the embodiment described above are indicated by the same reference numerals, and redundant descriptions are omitted.

First Variation of Fifth Embodiment:

[0102] FIG. **22** depicts a semiconductor device A**51** according to a first variation of the fifth embodiment. FIG. **22** is a sectional view of the semiconductor device A**51** and corresponds to FIG. **6**. The semiconductor device A**51** of the present variation does not include the positioning member **8**. In this variation, the positioning member **8** is removed during the manufacture of the semiconductor device A**51**. Thus, the semiconductor device A**51** is identical in configuration to the semiconductor device A**10** according to the first embodiment. The method for manufacturing the semiconductor device A**51** is identical to that for the semiconductor device A**50** of the fifth embodiment, from the reflowing step to the step of solidifying the solder paste **60**. The method for manufacturing the semiconductor device A**51** includes, as a subsequent step, removing the positioning member **8** before the step of bonding wire **65**. In the present embodiment, the positioning member **8** is made of a thermoplastic resin. Examples of the thermoplastic resin include polyethylene and polypropylene. The step of removing the positioning member **8** involves dissolving the positioning member **8** with an organic solvent. Alternatively, the positioning member **8** may be removed by other methods.

[0103] According to this variation, the semiconductor device A**51** includes the positioning member **8** when the conductive member **5** is bonded to the first electrode **21** and the pad portion **14**. Similarly to the semiconductor device A**50**, the semiconductor device A**51** therefore ensures precise control over the height of the conductive member **5** from the obverse surface **121**. According to the variation, however, the finished semiconductor device A**51** does not include the positioning member **8**. Thus, the semiconductor device A**51** is without the possibility of any gaps between the positioning member **8** and the sealing resin **7**. Consequently, the semiconductor device A**51** eliminates the risk of cracks originating from such a gap between the positioning member **8** and the sealing resin **7**.

[0104] Although the positioning member **8** in this variation has been described as being made of thermoplastic resin, this is not a limitation. The positioning member **8** may be made of a water soluble resin. Examples of the water soluble resin include polyethylene oxide, polyvinyl alcohol, resol-type phenolic resin, methylolated urea resin, methylolated melamine resin, polyacrylamide, and carboxymethyl cellulose. In this case, the step of removing the positioning member **8** involves dissolving the positioning member **8** with water.

Second Variation of Fifth Embodiment:

[0105] FIGS. **23** to **26** are views for illustrating a semiconductor device A**52** according to a second variation of the fifth embodiment. FIG. **23** is a sectional view of the semiconductor device A**52** and corresponds to FIG. **6**. FIGS. **24** to **26** are sectional views each illustrating a step of an example method for manufacturing the semiconductor device A**52**. The semiconductor device A**52** differs from the semiconductor device A**50** in the shapes of the positioning member **8** and the conductive member **5**.

[0106] According to the present variation, the conductive member **5** additionally includes a protruding portion **54**. The protruding portion **54** protrudes from the end surface **513** of the element bonding portion **51** toward the first side in the first direction x. The protruding portion **54** has a second reverse surface **542** facing the same side as the reverse surface **512** in the thickness direction z (the second side in the thickness direction z).

[0107] The positioning member **8** of the present variation does not include the second portion **82** and composed only of the first portion **81** that is a plate extending in the thickness direction z. The positioning member **8** is bonded to the obverse surface **121** of the die pad **12** at the end surface **81***a* facing the second side in the thickness direction z. The process for the bonding is not limited, and examples of the process include thermal compression bonding, which involves heating the positioning member **8** and the die pad **12** (the lead frame **100**) and pressing them together to form the bond. Alternatively, the positioning member **8** may be formed in contact with the obverse surface **121** of the die pad **12** by introducing molten resin material into a mold and then allowing it to harden. The positioning member **8** is also in contact with the second reverse surface **542** of the protruding portion **54** of the conductive member **5** at the end surface **81***b* facing the first side in the thickness direction z. In other words, the positioning member **8** is in contact with the obverse surface **121** of the die pad **12** and the second reverse surface **542** of the conductive member **5**.

[0108] The following describes the example method for manufacturing the semiconductor device A**52**, with reference to FIGS. **24** to **26**. FIGS. **24** to **26** are sectional views each illustrating a step of

the method for manufacturing the semiconductor device A**52** and correspond to FIG. **6**.
[0109] First, a lead frame **100** and a positioning member **8** are prepared as shown in FIG. **24**. Subsequently, as shown in FIG. **24**, the positioning member **8** is joined by thermal compression bonding to the portion of the obverse surface **101** of the lead frame **100** that is for forming the obverse surface **121** of the die pad **12**. As a result, the positioning member **8** and the die pad **12** (the lead frame **100**) are joined together into a single unit. Alternatively, the positioning member **8** may be formed in contact with the obverse surface **101** of the lead frame **100** by placing the lead frame **100** into a mold, introducing molten resin material into the mold, and then allowing it to harden. In a separate step, the conductive member **5** and the semiconductor element **2** are prepared.
[0110] Subsequently, solder paste **60** is applied to the obverse surface **101** of the lead frame **100** to cover the portion for forming the obverse surface **121** of the die pad **12**, and the semiconductor element **2** is placed on the applied solder paste **60**. Subsequently, as shown in FIG. **25**, the solder paste **60** is applied to the first electrode **21** of the semiconductor element **2**.
[0111] Subsequently, as shown in FIG. **26**, the conductive member **5** is placed to span across the semiconductor element **2** and the portion of the lead frame **100** that is for forming the pad portion **14**. In this state, the element bonding portion **51** is placed on the solder paste **60**, whereas the protruding portion **521** of the lead bonding portion **52** is placed directly on the portion of the lead frame **100** that is for forming the pad portion **14**. The conductive member **5** is placed such that the second reverse surface **542** of the protruding portion **54** is in contact with the end surface **81***b* of the positioning member **8**, which has been integrated with the lead frame **100**. The subsequent steps are the same as those for the method for manufacturing the semiconductor device A**50** according to the fifth embodiment.
[0112] According to the present variation, the semiconductor device A**52** includes the positioning member **8**. The positioning member **8** is integrated with the die pad **12** by bonding the end surface **81***a* directly to the obverse surface **121** of the die pad **12**. The conductive member **5** is secured in the state where the second reverse surface **542** of the protruding portion **54** is in contact with the end surface **81***b* of the positioning member **8**, which is integral with the die pad **12** (the lead frame **100**). Thus, the height (the position in the thickness direction z) of the conductive member **5** from the obverse surface **121** is determined by the dimension of the positioning member **8** in the thickness direction z. Similarly to the semiconductor device A**50**, the semiconductor device A**52** of the present variation therefore ensures precise control over the height of the conductive member **5** from the obverse surface **121**. According to the present variation, in addition, the positioning member **8** extends in the thickness direction z. Thus, the positioning member **8** is placed such that the end surface **81***b* is in contact with the second reverse surface **542** of the protruding portion **54**, and that the end surface **81***a* is in contact with the obverse surface **121** of the die pad **12**.
[0113] Although the present variation described above is an example in which the positioning member **8** is joined to the die pad **12** first. However, this is not a limitation. For instance, it is possible to join the end surface **81***b* of the positioning member **8** to the second reverse surface **542** of the protruding portion **54** before the positioning member **8** is joined to the conductive member **5**. It is also possible that the conductive member **5** does not include the protruding portion **54**. Instead, the element bonding portion **51** has a portion extending beyond the first side of the semiconductor element **2** in the first direction, and its the reverse surface **512** is in contact with the positioning member **8**.
[0114] The semiconductor devices and methods for manufacturing semiconductor devices are not limited to those described above in the embodiments. Various modifications in design may be made freely in the specific structure of each part of the semiconductor devices and the steps of the method for manufacturing semiconductor devices according to the present disclosure.
[0115] The present disclosure includes embodiments described in the following clauses.
Clause 1.
[0116] A semiconductor device (A**1**) comprising: [0117] a first lead (**1**A) including a die pad (**12**)

that includes a die-pad obverse surface (**121**) facing a first side in a thickness direction (z); [0118] a semiconductor element (**2**) including an element obverse surface (**201**) facing the first side in the thickness direction, and a first electrode (**21**) disposed on the element obverse surface, the semiconductor element being mounted on the die-pad obverse surface; [0119] a second lead (**1**B) including a second obverse surface (**141**) facing the first side in the thickness direction, the second lead being spaced apart from the first lead in a first direction (x) perpendicular to the thickness direction; [0120] a conductive member (**5**) electrically bonded to the first electrode and the second obverse surface; and [0121] a sealing resin (**7**) covering the semiconductor element, [0122] wherein the conductive member is in direct contact with the second lead.
Clause 2.
[0123] The semiconductor device according to Clause 1, wherein a solid-state bonding interface (**59**) is present between the conductive member and the second lead.
Clause 3. (Second Embodiment, FIG. **14**)
[0124] The semiconductor device according to Clause 1, wherein the conductive member includes a welding mark (**523**) that extends to an interior of the second lead.
Clause 4.
[0125] The semiconductor device according to any one of Clauses 1 to 3, wherein the conductive member and the second lead are made of a same constituent material.
Clause 5.
[0126] The semiconductor device according to any one of Clauses 1 to 4, wherein a constituent material of the conductive member and the second lead contains Cu.
Clause 6.
[0127] The semiconductor device according to any one of Clauses 1 to 5, wherein the second lead includes a pad portion (**14**) covered with the sealing resin, and a terminal portion (**15**) including a portion exposed from the sealing resin, and [0128] the second obverse surface is located in the pad portion.
Clause 7.
[0129] The semiconductor device according to any one of Clauses 1 to 6, wherein the conductive member includes a conductive-member obverse surface (**511**) facing the first side in the thickness direction, and [0130] the conductive-member obverse surface is exposed from the sealing resin.
Clause 8. (Third Embodiment, FIG. **15**)
[0131] The semiconductor device according to any one of Clauses 1 to 6, further comprising a heat-conducting member (**9**) bonded to the conductive member, [0132] wherein the conductive member includes a conductive-member obverse surface facing the first side in the thickness direction, and [0133] the heat-conducting member is bonded to the conductive-member obverse surface and is exposed from the sealing resin.
Clause 9.
[0134] The semiconductor device according to any one of Clauses 1 to 8, wherein the die pad further includes a die-pad reverse surface (**122**) facing a second side in the thickness direction, and [0135] the die-pad reverse surface is exposed from the sealing resin.
Clause 10. (Fifth Embodiment, FIGS. **17** to **26**)
[0136] The semiconductor device according to any one of Clauses 1 to 9, further comprising a positioning member (**8**) containing an insulating material and in contact with the conductive member and the die-pad obverse surface.
Clause 11. (Fifth Embodiment, FIG. **17**)
[0137] The semiconductor device according to Clause 10, wherein the conductive member includes a conductive-member end surface (**513**) facing a first side in the first direction, and [0138] the positioning member includes a first portion (**81**) and a second portion (**82**), the first portion being in contact with the die-pad obverse surface and extending in the thickness direction, the second portion being in contact with the conductive-member end surface and extending in the first

direction.

Clause 12. (Second Variation of Fifth Embodiment, FIG. **23**)

[0139] The semiconductor device according to Clause 10, wherein the conductive member includes a conductive-member reverse surface (**542**) facing a second side in the thickness direction, and [0140] the positioning member is in contact with the die-pad obverse surface and the conductive-member reverse surface and extends in the thickness direction.

Clause 13. (FIGS. **10** to **12**)

[0141] A method for manufacturing a semiconductor device, the method comprising: [0142] placing a semiconductor element on a first bonding member (**60**) located on a die-pad obverse surface of a die pad; [0143] placing a second bonding member (**60**) on a first electrode of the semiconductor element; [0144] placing a conductive member to span across the semiconductor element and a second lead that is spaced apart from the die pad; [0145] bonding the conductive member and the second lead in direct contact with each other; and [0146] heating to solidify the first bonding member and the second bonding member.

Clause 14.

[0147] The method according to Clause 13, wherein the bonding involves bonding the conductive member and the second lead by ultrasonic bonding.

Clause 15.

[0148] The method according to Clause 13, wherein the bonding involves bonding the conductive member and the second lead by laser welding.

REFERENCE NUMERALS

[0149] A**10** to A**11**, A**20**, A**30**, A**40**, A**50** to A**52**: semiconductor device [0150] **1**A, **1**B, **1**C: lead **12**: die pad **121**: obverse surface **122**: reverse surface [0151] **13**: first terminal portion **131**: reverse-surface mounting portion [0152] **14**: pad portion **141**: obverse surface **142**: plating layer **15**: second terminal portion [0153] **151**: reverse-surface mounting portion **16**: bent portion **17**: pad portion [0154] **18**: second terminal portion **181**: reverse-surface mounting portion **19**: bent portion [0155] **2**: semiconductor element **20**: element body **201**: element obverse surface [0156] **202**: element reverse surface **21**: first electrode **212**: first-electrode pad portion [0157] **22**: second electrode [0158] **5**: conductive member [0159] **23**: third electrode **3**: insulating part **4**: metal laminate [0160] **51**: element bonding portion **511**: obverse surface [0161] **512**: reverse surface **513**: end surface **52**: lead bonding portion [0162] **521**: protruding portion **521***a*: contact surface **522**: plating layer **523**: welding mark [0163] **53**: intermediate portion **54**: protruding portion **542**: second reverse surface [0164] **59**: solid-state bonding interface **61**, **62**: conductive bonding material **65**: wire [0165] **7**: sealing resin **71**: resin obverse surface **72**: resin reverse surface [0166] **73**, **74**, **75**, **76**: resin side surface **8**: positioning member **81**: first portion [0167] **81***a*, **81***b*: end surface **82**: second portion **82***a*: end surface [0168] **9**: heat-conducting member **91**: obverse surface **92**: reverse surface [0169] **9***a*: insulating plate **9***b*: metal layer **100**: lead frame [0170] **101**: obverse surface **60**: solder paste

Claims

1. A semiconductor device comprising: a first lead including a die pad that includes a die-pad obverse surface facing a first side in a thickness direction; a semiconductor element including an element obverse surface facing the first side in the thickness direction, and a first electrode disposed on the element obverse surface, the semiconductor element being mounted on the die-pad obverse surface; a second lead including a second obverse surface facing the first side in the thickness direction, the second lead being spaced apart from the first lead in a first direction perpendicular to the thickness direction; a conductive member electrically bonded to the first electrode and the second obverse surface; and a sealing resin covering the semiconductor element, wherein the conductive member is in direct contact with the second lead.

2. The semiconductor device according to claim 1, wherein a solid-state bonding interface is

present between the conductive member and the second lead.
3. The semiconductor device according to claim 1, wherein the conductive member includes a welding mark that extends to an interior of the second lead.
4. The semiconductor device according to claim 1, wherein the conductive member and the second lead are made of a same constituent material.
5. The semiconductor device according to claim 1, wherein a constituent material of the conductive member and the second lead contains Cu.
6. The semiconductor device according to claim 1, wherein the second lead includes a pad portion covered with the sealing resin, and a terminal portion including a portion exposed from the sealing resin, and the second obverse surface is located in the pad portion.
7. The semiconductor device according to claim 1, wherein the conductive member includes a conductive-member obverse surface facing the first side in the thickness direction, and the conductive-member obverse surface is exposed from the sealing resin.
8. The semiconductor device according to claim 1, further comprising a heat-conducting member bonded to the conductive member, wherein the conductive member includes a conductive-member obverse surface facing the first side in the thickness direction, and the heat-conducting member is bonded to the conductive-member obverse surface and is exposed from the sealing resin.
9. The semiconductor device according to claim 1, wherein the die pad further includes a die-pad reverse surface facing a second side in the thickness direction, and the die-pad reverse surface is exposed from the sealing resin.
10. The semiconductor device according to claim 1, further comprising a positioning member containing an insulating material and in contact with the conductive member and the die-pad obverse surface.
11. The semiconductor device according to claim 10, wherein the conductive member includes a conductive-member end surface facing a first side in the first direction, and the positioning member includes a first portion and a second portion, the first portion being in contact with the die-pad obverse surface and extending in the thickness direction, the second portion being in contact with the conductive-member end surface and extending in the first direction.
12. The semiconductor device according to claim 10, wherein the conductive member includes a conductive-member reverse surface facing a second side in the thickness direction, and the positioning member is in contact with the die-pad obverse surface and the conductive-member reverse surface and extends in the thickness direction.
13. A method for manufacturing a semiconductor device, the method comprising: placing a semiconductor element on a first bonding member located on a die-pad obverse surface of a die pad; placing a second bonding member on a first electrode of the semiconductor element; placing a conductive member to span across the semiconductor element and a second lead that is spaced apart from the die pad; bonding the conductive member and the second lead in direct contact with each other; and heating to solidify the first bonding member and the second bonding member.
14. The method according to claim 13, wherein the bonding involves bonding the conductive member and the second lead by ultrasonic bonding.
15. The method according to claim 13, wherein the bonding involves bonding the conductive member and the second lead by laser welding.